



Technological and Nutritional Quality Traits and Globulin Protein Polymorphism in Chickpea (*Cicer arietinum* L.) Germplasm

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ABSTRACT

Background: Chickpea (*Cicer arietinum* L.) is an important grain legume crop with high nutritional value due to its protein content, essential amino acids and functional properties. Genetic variation among chickpea germplasm collections provides valuable opportunities for improving seed nutritional quality and identifying promising materials for breeding programs. However, comprehensive evaluation combining technological traits, biochemical composition and storage-protein polymorphism remains limited for many chickpea collections. The present study aimed to characterize the variation in technological quality traits, nutritional composition and globulin storage-protein profiles among chickpea accessions to identify potential breeding materials with improved seed quality.

Methods: Field and laboratory experiments were conducted during 2023–2025 at the Absheron Experimental Base Station of the Institute of Genetic Resources, Ministry of Science and Education of the Republic of Azerbaijan. A total of 51 chickpea accessions, including local cultivars and introduced germplasm, were evaluated. Technological traits, including water absorption capacity, cooking time, moisture content and 100-seed weight, were assessed. Biochemical analyses were performed to determine total protein, lysine and tryptophan contents. Globulin storage-protein polymorphism was investigated using acid polyacrylamide gel electrophoresis (A-PAGE). Statistical analyses included ANOVA, Pearson correlation analysis and principal component analysis (PCA).

Result: Significant variation was observed among chickpea accessions for technological and biochemical traits. The 100-seed weight ranged from 27.5 to 43.9 g, while total protein content varied from 21.3% to 27.2%. Several accessions demonstrated high nutritional potential, including Filip 08-65, which showed the highest lysine and tryptophan contents. Correlation analysis revealed relationships among protein, lysine and tryptophan contents, while PCA differentiated accessions according to their biochemical characteristics. Electrophoretic analysis of globulin storage proteins revealed considerable polymorphism among the studied genotypes, with the highest diversity observed in the β region. The identified accessions may serve as valuable genetic resources for future chickpea breeding programs targeting improved nutritional quality and storage-protein diversity.

Key words: Chickpea, Storage-protein polymorphism, Protein content, Lysine, Tryptophan, Seed proteins, Nutritional quality, Germplasm.

INTRODUCTION

Cicer arietinum L. (chickpea) is an annual, predominantly self-pollinating pulse crop belonging to the family Fabaceae (Vélez *et al.*, 2025). It is widely accepted that chickpea originated and underwent primary domestication in the region encompassing present-day southeastern Türkiye and northern Syria. Subsequent diversification is believed to have occurred across the Middle East, Central Asia, the Indian subcontinent and the Mediterranean basin (Igolkina *et al.*, 2023; Zhang *et al.*, 2024). Currently cultivated in more than 50 countries worldwide, chickpea represents one of the most important grain legumes and is ranked as the third most widely produced food legume globally, following dry beans and dry peas (Zhang *et al.*, 2024).

The high content of proteins, amino acids, vitamins, minerals and bioactive compounds in chickpea seeds has made chickpea an important plant-based food source in human nutrition (Kaur and Prasad, 2021; Begum *et al.*, 2023; Jha *et al.*, 2024; Yadav *et al.*, 2025).

Cicer arietinum has attracted considerable attention not only because of its high nutritional value but also

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because of its functional and potential health-promoting properties. Bioactive compounds present in chickpea have been associated with antioxidant and health-promoting properties (Begum *et al.*, 2023; Jha *et al.*, 2024). Furthermore, experimental evidence suggests potential effects on glycaemic regulation, lipid metabolism and gut health (Faridy *et al.*, 2020; Begum *et al.*, 2023).

The seed proteins of *Cicer arietinum* L. consist predominantly of globulins and albumins, which account for approximately 55-60% and 8-14% of total seed protein, respectively. Globulins represent the principal storage protein fraction, whereas albumins comprise mainly water-soluble proteins with enzymatic and metabolic functions. In addition, albumin fractions may contain antinutritional compounds, including trypsin inhibitors (Zha *et al.*, 2021; Grasso *et al.*, 2022). Chickpea proteins are considered valuable sources of essential amino acids (Ipekessen *et al.*, 2022; Begum *et al.*, 2023). Tryptophan is generally present at relatively low concentrations and may not be detected in some analyses because of methodological limitations or detection thresholds. However, other studies have identified chickpea genotypes characterised by elevated tryptophan contents (Salmanova *et al.*, 2026). Tryptophan plays an important role in neuronal function, immune regulation and gut homeostasis (Luo *et al.*, 2026).

Chickpea is a nutrient-dense and affordable source of plant-based protein with potential to support food security in developing regions. Genotype-dependent variation in biochemical composition affects its nutritional and functional properties and may contribute to stress adaptation, defence responses and metabolic regulation (Thudi *et al.*, 2017; Rajput *et al.*, 2023). Due to its low production requirements, adaptive capacity, extensive root system and ability to fix atmospheric nitrogen through symbiosis, chickpea is also an important component of sustainable agricultural systems, improving soil fertility and reducing reliance on mineral fertilizers (Mohsenzadeh, 2024). Therefore, this study aimed to characterize technological traits, biochemical composition and globulin storage-protein profiles of local and introduced chickpea genotypes and to identify promising accessions for breeding programmes targeting improved nutritional quality.

MATERIALS AND METHODS

Plant material

The field and laboratory experiments were conducted from 2023 to 2025 at the Absheron Experimental Base Station of the Institute of Genetic Resources, Ministry of Science and Education of the Republic of Azerbaijan. The research site is located on the Caspian Sea coast, southeast of Baku, at 40.413010°N and 49.833421°E.

The Absheron Peninsula is characterized by a dry subtropical climate. Mean annual air temperature ranges from 12 to 15°C; minimum temperatures during winter

generally range from -2 to 4°C, whereas maximum temperatures in summer reach 28-35°C. Mean daily air temperature remains above 0°C. Annual precipitation is approximately 200-250 mm. These climatic conditions are considered favourable for the cultivation of legume crops. The investigated chickpea accessions were sown under open-field conditions during the autumn season at the experimental site of the Genetic Resources Institute. Seeds from 51 mature chickpea accessions were used as the experimental material. The local cultivar Cemile and Cemile 1 originated from Azerbaijan and was maintained in the chickpea germplasm collection of the Institute of Genetic Resources. The remaining 49 chickpea accessions were introduced into Azerbaijan from the International Center for Agricultural Research in the Dry Areas (ICARDA) and subsequently maintained at the Institute's Absheron Experimental Base Station (Supplementary Table S1).

Biochemical analyses

Determination of protein content

Total protein content in chickpea grains was determined using the Kjeldahl method. 100 mg of finely ground chickpea flour was transferred to a 100 mL Kjeldahl digestion flask, followed by the addition of 7 mL of concentrated sulfuric acid (H₂SO₄). A selenium-copper mixture was used as the digestion catalyst.

The samples were digested in a fume hood until a clear solution was obtained. After cooling, the inner walls of the flask were rinsed with distilled water and the samples were heated for an additional 10 min to ensure complete digestion. Subsequently, the digested samples were subjected to distillation using a Kjeldahl apparatus.

For ammonia collection, 25 mL of 0.02 N H₂SO₄ and 1-2 drops of indicator were added to each 200 mL receiving flask. The digestion flasks were connected to the Kjeldahl apparatus and the digested samples were alkalized with a 40% alkaline solution. Distillation was continued for 20-30 min. The solution collected in the receiving flask was then back-titrated with 0.02 N NaOH until the endpoint, indicated by a distinct color change, was reached.

The total protein content was calculated according to the following equation:

$$\text{Protein content (\%)} = \frac{(25 - V) \times 0.28 \times 100}{m} \times \frac{100}{89} \times 6.25$$

Where

V = Volume of 0.02 N NaOH used for back-titration (mL).

25 = Initial volume of 0.02 N H₂SO₄ added to the receiving flask (mL).

0.28 = Nitrogen-equivalent coefficient for 0.02 N H₂SO₄.

m = Mass of the analyzed sample; 100/89 is the dry-matter correction factor.

6.25 = Nitrogen-to-protein conversion factor (Vinklářková, 2015).

Determination of lysine content

Lysine content was determined in duplicate using the ninhydrin-based colorimetric method described by Museyko and Sysoev (1970), with minor modifications. Briefly, 30 mg of finely ground chickpea flour was transferred to a 100 mL flask, followed by the addition of 1 mL of 2% Na_2CO_3 solution and 250 mg of finely powdered glass. The mixture was thoroughly homogenized with a glass rod and incubated in a water bath at 80°C for 10 min.

After the initial incubation, 2 mL of ninhydrin reagent was added to each sample. Calibration standards were prepared by mixing 0.5 mL of each L-lysine standard solution (0-100 µg/mL L-lysine) with 0.5 mL of 4% Na_2CO_3 solution and 2 mL of ninhydrin reagent. A reagent blank containing all reagents but no chickpea sample or L-lysine standard was prepared in parallel. The samples, standards and reagent blank were incubated in a water bath for 30 min under identical conditions.

Following colour development, the mixtures were cooled to room temperature, supplemented with 5 mL of 95% ethanol, thoroughly mixed and centrifuged at for 5 min. The absorbance of the clear supernatant was measured against the reagent blank using a photoelectric colorimeter (Ultrospec™ 2100 pro UV/Vis spectrophotometer (Amersham Biosciences, UK). Lysine concentrations were determined from the linear calibration curve constructed using the L-lysine standards. The calibration equation was expressed as:

$$A=aC+b$$

Where

A = Blank-corrected absorbance.

C = L-lysine concentration.

a = Slope.

b = Intercept.

Calibration performance was evaluated using the coefficient of determination.

Lysine content was calculated according to the following equation:

$$\text{Lysine (mg/100 g DW)} = mC \times V \times DF \times 100$$

Where

C= Lysine concentration obtained from the calibration curve (mg/mL).

V = Final extract volume (mL).

DF = Dilution factor.

m = Dry mass of the analysed sample (g).

The results were expressed as mg/100 g dry weight (DW).

Determination of tryptophan content

Tryptophan content in synthetic chickpea grains was determined according to the colorimetric method of Ermakov and Yarosh (1969), with minor modifications. Briefly, 200 mg of finely ground chickpea flour was transferred into a 100 mL volumetric flask and 2 mL of distilled water was added. Subsequently, 1 mL of a 4% gelatin solution prepared in 25% KOH was added. The flasks

were tightly sealed with rubber stoppers and the samples were incubated at 40°C for 18-20 h (Ermakov and Yarosh, 1969).

Following incubation, the samples were cooled to room temperature. Subsequently, 0.5 mL of p-dimethyl aminobenzaldehyde reagent prepared in 10% HCl, 0.5 mL of 1% sodium nitrate (NaNO_3) solution and 28 mL of concentrated HCl were sequentially added to each flask. The reaction mixtures were thoroughly mixed, resealed and incubated in a thermostatically controlled chamber at 25°C for 90 min.

After color development, the volume of each sample was adjusted to 100 mL with distilled water. The resulting solutions were filtered and their absorbance was measured at 590 nm against a reagent blank using a spectrophotometer (Amersham Biosciences, UK). Tryptophan concentration was determined using an external calibration curve prepared from analytical-grade L-tryptophan standards subjected to the same color-development procedure.

Tryptophan content was calculated according to the following equation:

$$\text{Tryptophan content (mg 100 g}^{-1}\text{)} = \frac{C \times V \times DF \times 100}{m}$$

Where

C = Tryptophan concentration obtained from the calibration curve (mg mL⁻¹).

V = Final volume of the sample solution (mL).

DF = Dilution factor.

m = Mass of the analyzed grain sample (g).

The results were expressed as milligrams of tryptophan per 100 g of grain dry matter.

Electrophoretic analysis (A-PAGE)

Globulin storage proteins from 51 chickpea genotypes were extracted and analysed by acid polyacrylamide gel electrophoresis (A-PAGE) using a modified Poperelya (1989) protocol. Since the original method was developed for wheat gliadin analysis, adaptations for chickpea storage proteins included two preliminary extractions with 70% ethanol, two washes with 0.03% acetic acid-acetone solution, incubation for 15 min and centrifugation at 3500 rpm for 5 min after each step, followed by solubilization of the pellet in 500 µL of 9 M urea-acetic acid solution. Protein extracts were subsequently separated in a vertical A-PAGE system using glycine-acetate buffer (pH 3.5).

These modifications were introduced to improve extraction and electrophoretic resolution of chickpea globulin storage proteins while maintaining the original principles of the Poperelya method.

For consistent band scoring, each electrophoretic profile was divided into four operational mobility regions (Regions I-IV), arranged from the upper to the lower part of the gel. These regions were defined solely according to relative electrophoretic mobility and were used to facilitate consistent comparison of banding patterns among chickpea

Supplementary Table S1: Technological and seed quality characteristics of local and introduced chickpea genotypes.

Accession name	Origin	Status	Water absorption capacity (mL)	Cooking time (min)	Seed colour	Moisture (%)	100-seed weight (g)
Camila	Azerbaijan	Local cultivar	15	29	Yellow	13.6	36.0
Filip 08-57	ICARDA	Introduced accession	14	30	Yellow	12.6	41.0
Filip 05-43	ICARDA	Introduced accession	11	34	Yellow	11.8	34.8
Filip 08-27	ICARDA	Introduced accession	12	35	Yellow	11.6	33.9
Filip 07-104	ICARDA	Introduced accession	12	30	Yellow	12.8	37.8
Filip 07-115	ICARDA	Introduced accession	15	29	Yellow	13.2	42.9
Filip 08-65	ICARDA	Introduced accession	13	32	Yellow	12.0	38.5
Filip 07-91	ICARDA	Introduced accession	11	35	Yellow	11.0	33.5
Camila 1	Azerbaijan	Local cultivar	15	29	Yellow	13.6	36.0
Filip 08-44	ICARDA	Introduced accession	13	36	Yellow	11.5	38.6
Filip 08-58	ICARDA	Introduced accession	11	35	Yellow	10.0	28.9
AC 3190	ICARDA	Introduced accession	11	35	Black	11.6	27.5
AC 3199	ICARDA	Introduced accession	14	30	Light yellow	12.8	41.7
Filip 06-69	ICARDA	Introduced accession	12	35	Light yellow	10.8	30.8
Filip 88-85	ICARDA	Introduced accession	15	29	Yellow	13.6	43.9
Filip 93-93	ICARDA	Introduced accession	15	29	Dark yellow	13.5	28.9
Filip 13-234C	ICARDA	Introduced accession	12	34	Yellow	12.0	28.6
Filip 08-13	ICARDA	Introduced accession	10	38	Yellow	10.0	32.9
Filip 06-54	ICARDA	Introduced accession	12	36	Dark yellow	11.7	35.0
Filip 05-74	ICARDA	Introduced accession	10	37	Yellow	10.8	34.9
Filip 07-74	ICARDA	Introduced accession	11	35	Yellow	10.6	34.9
Filip 09-303	ICARDA	Introduced accession	15	29	Yellow	13.2	36.0
Filip 09-229	ICARDA	Introduced accession	12	35	Yellow	11.6	35.0
Filip 09-240	ICARDA	Introduced accession	15	30	Dark yellow	13.7	35.3
Filip 09-218	ICARDA	Introduced accession	13	30	Yellow	12.6	32.5
Filip 09-135	ICARDA	Introduced accession	12	36	Yellow	11.6	34.0
Filip 09-204	ICARDA	Introduced accession	12	36	Yellow	13.8	35.5
Filip 13-138	ICARDA	Introduced accession	13	33	Yellow	12.6	38.6
Filip 09-285	ICARDA	Introduced accession	12	35	Dark yellow	11.5	32.6
Filip 11-08	ICARDA	Introduced accession	13	30	Yellow	12.0	32.7
Filip 09-94	ICARDA	Introduced accession	12	35	Dark yellow	11.6	34.2
Filip 09-291	ICARDA	Introduced accession	13	33	Yellow	12.0	38.0
Filip 09-96	ICARDA	Introduced accession	11	37	Yellow	11.3	30.2
Filip 09-293	ICARDA	Introduced accession	11	32	Black	11.6	34.0
Filip 09-306	ICARDA	Introduced accession	8	38	Dark yellow	10.0	30.0
Filip 09-37	ICARDA	Introduced accession	11	32	Yellow	10.8	30.0
Filip 03-140	ICARDA	Introduced accession	14	30	Light yellow	13.0	33.4
Filip 98-117	ICARDA	Introduced accession	10	35	Light yellow	10.0	29.8
Filip 06-37	ICARDA	Introduced accession	12	33	Dark yellow	12.0	36.5
Filip 05-9	ICARDA	Introduced accession	14	35	Yellow	11.8	38.6
Filip 07-183	ICARDA	Introduced accession	15	29	Yellow	13.0	41.8
Filip 07/H 127	ICARDA	Introduced accession	15	29	Yellow	13.5	40.5
Filip 06-114	ICARDA	Introduced accession	15	30	Yellow	12.8	41.5
Filip 05-83	ICARDA	Introduced accession	14	30	Yellow	12.9	38.0
Filip 06-69	ICARDA	Introduced accession	10	36	Light yellow	10.6	33.8
Filip 06-141	ICARDA	Introduced accession	15	30	Black	12.8	32.9
Filip 07-201	ICARDA	Introduced accession	10	37	Yellow	10.0	34.0
Filip 09-30	ICARDA	Introduced accession	10	35	Yellow	11.6	32.0
Filip 02-28	ICARDA	Introduced accession	11	35	Dark yellow	11.8	28.4
Filip 05-36	ICARDA	Introduced accession	12	33	Dark yellow	12.0	36.2
Filip 05-59	ICARDA	Introduced accession	12	33	Yellow	12.3	36.0

accessions; they were not interpreted as biochemically homologous to the α -, β -, γ -, or ω -gliadin fractions of wheat (Bushuk and Zillman, 1978). For each operational mobility region, an electrophoretic pattern class was defined as a unique combination of reproducible presence/absence bands. Accessions with identical band combinations were assigned to the same pattern class, while any reproducible difference in at least one band defined a different class. Pattern diversity was calculated as:

$$H = 1 - \sum P_i^2$$

Where

H = Storage-protein pattern diversity index.

P_i = Frequency of the i^{th} electrophoretic pattern within each zone.

Because the input data represent electrophoretic protein-band phenotypes rather than directly genotyped loci, H was interpreted as an index of storage-protein pattern diversity and not as a genome-wide estimate of storage-protein pattern diversity.

Cluster analysis of electrophoretic profiles

Only clear and reproducible globulin bands were used for cluster analysis. Pairwise dissimilarities among accessions were calculated using the Jaccard coefficient was constructed using DARwin software version 6.0. As no bootstrap analysis was performed, the clustering was considered an exploratory representation of variation in globulin storage-protein profiles rather than statistically supported genetic relationships.

Statistical analysis

All technological and biochemical traits were evaluated in three replicates for each of the 51 chickpea accessions and results were expressed as mean $\pm$ standard deviation (SD). Differences among accessions were analyzed using one-way ANOVA, with accession as the fixed factor. Variance homogeneity was assessed using Levene's test and significant differences were further evaluated by Tukey's HSD post-hoc test at $P < 0.05$. In tables, means sharing the same lowercase letter indicate no significant difference according to Tukey's HSD test.

Pearson's correlation analysis was performed using accession mean values ($n = 51$) to assess relationships among water absorption capacity, cooking time, moisture content, 100-seed weight, protein, lysine and tryptophan contents. Correlation significance was tested using two-tailed tests at $P < 0.05$, $P < 0.01$ and $P < 0.001$.

Principal component analysis (PCA) was conducted using standardized accession means (z-score transformation) based on the correlation matrix. Principal components were interpreted according to eigenvalues, explained variance and trait loadings. Cluster analysis of technological traits was visualized using a heat map based on Euclidean distance and Ward's hierarchical clustering method.

RESULTS AND DISCUSSION

Chickpea is regarded as an important food source among grain legumes because of its high nutritional value. To evaluate variation in nutritional quality traits among the studied chickpea genotypes, a range of technological and biochemical analyses was performed on their seeds. The results of these analyses are presented in Fig 1. There is a strong negative relationship between water absorption capacity and cooking time and a strong positive relationship between absorbability and moisture content. The results showed that, among the 51 chickpea accessions evaluated, 12 exhibited superior technological characteristics (Fig 1).

Based on the results of the study, the highest 100-seed weight values were recorded in five chickpea accessions: Filip 07/H127 (40.5 g), Filip 06-114 (41.5 g), Filip 07-183 (41.8 g), Filip 07-115 (42.9 g) and Filip 88-85 (43.9 g). These values were 4.5-8.4 g higher than that of the standard cultivar Cemile, corresponding to an increase of 12.5-23.3% in 100-seed weight. The considerable variation observed in 100-seed weight among the evaluated accessions agrees with previous studies reporting significant genotypic variability for this trait in chickpea. Moreover, high heritability coupled with high genetic advance has been reported for 100-seed weight, indicating that this trait may respond effectively to selection (Dehal *et al.*, 2016; Yücel, 2020). Accessions exhibiting comparatively favourable technological characteristics generally showed water absorption capacities of 14-15 mL and cooking times of 29-30 min. However, because no predefined composite selection index was applied, the accessions were evaluated separately for each technological trait rather than classified into a single overall "superior" group.

The biochemical composition of the accessions was evaluated based on protein, tryptophan and lysine contents (Fig 2).

For descriptive screening, comparatively high protein content was operationally defined as $\geq 25.6\%$ DW. These values represented relative increases of approximately 6.7-13.3% compared with Cemile (24.0%), equivalent to differences of 1.6-3.2 percentage points. Although these accessions showed higher mean values, Tukey's HSD indicated no significant differences from Cemile. The overall effect of accession on protein content was significant ($P < 0.001$), Tukey's HSD test showed that none of these nine accessions differed significantly from Cemile ($P > 0.05$). Thus, their higher mean protein contents should be interpreted as descriptive differences and potential candidates for further evaluation rather than statistically confirmed superiority over Cemile.

Tryptophan content in the evaluated chickpea accessions was lower than that of the standard cultivar Cemile. However, among the 51 accessions analyzed, 11 exhibited tryptophan contents ranging from 210 to 250 mg/100 g dry weight. The highest tryptophan content (250 mg/100 g dry weight) was recorded in the Filip 08-27, Filip 08-65 and Filip 08-58 accessions. In a comparable study, the

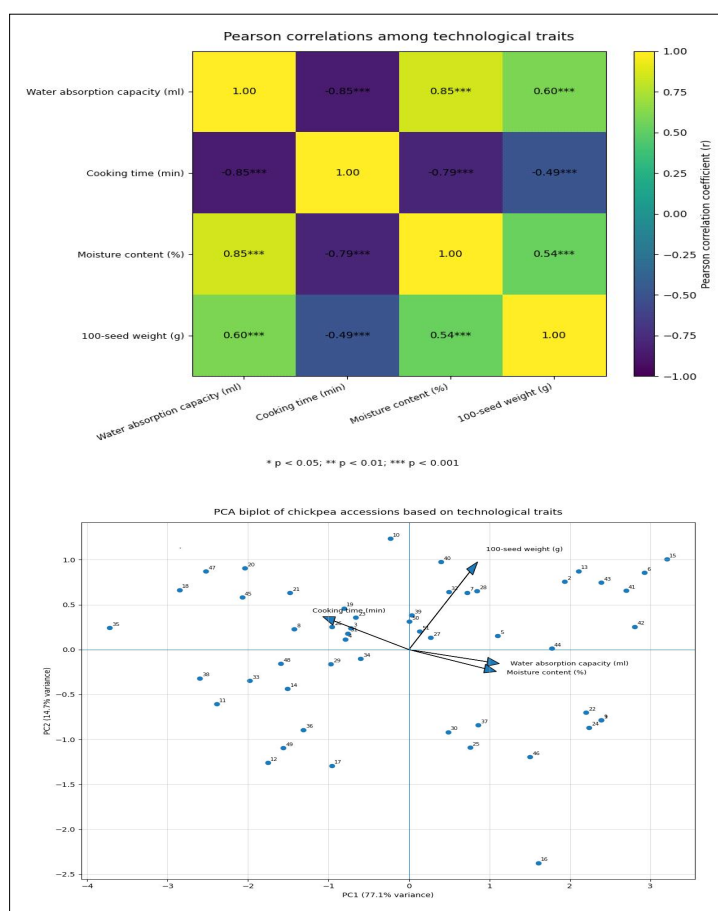


Fig 1: Pearson's correlation and PCA biplot analysis among technological quality traits of chickpea (*Cicer arietinum* L.).

tryptophan content of chickpea seeds was reported as 0.44 ± 0.03 mg g⁻¹ on a dry-weight basis. Among several legume crops investigated, including soybean, common bean, lentil and chickpea, chickpea was found to have the lowest tryptophan content (Köse *et al.*, 2024).

High lysine contents were identified in 12 chickpea accessions. These values were approximately twofold higher than those recorded in the standard cultivar, Cemile. The highest lysine content was observed in Filip 08-65, Filip 09-240, Filip 09-285 and Filip 06-114. Previous studies have also reported that lysine is among the most abundant essential amino acids in chickpea seeds relative to other amino acids (Ipekese *et al.*, 2022).

The relationships among protein, lysine and tryptophan contents were further evaluated using Pearson correlation analysis and principal component analysis (PCA). The correlation matrix revealed a significant positive association between protein and lysine contents, whereas tryptophan showed a weaker relationship with the other biochemical traits. The PCA analysis demonstrated the distribution of chickpea accessions according to their biochemical characteristics (Fig 3).

Based on the biochemical analysis, Filip 08-65 was the only accession combining the highest observed lysine

content (1294 mg/100 g DW) with the highest tryptophan content (250 mg/100 g DW), although its mean protein content was comparatively moderate (25.6% DW). When expressed relative to protein content, these values corresponded to 50.5 mg lysine/g protein and 9.8 mg tryptophan/g protein, calculated by dividing the amino acid content expressed per 100 g DW by the protein content expressed as g/100 g DW. Filip 03-140 and Filip 09-240 were tied for the highest mean protein content (27.2% DW). Filip 09-240 also contained 1294 mg lysine/100 g DW, corresponding to 47.6 mg lysine/g protein. The second-highest tryptophan value was observed in Filip 07-115 (240 mg/100 g DW), corresponding to 9.2 mg/g protein. Although these results indicate comparatively favourable lysine and tryptophan concentrations in selected accessions, the overall adequacy of their protein relative to the FAO/WHO/UNU reference pattern cannot be established because the complete essential amino-acid profile and protein digestibility were not evaluated. Electrophoretic analysis of globulin storage proteins was performed in the seeds of 51 chickpea genotypes selected for the study using acid polyacrylamide gel electrophoresis (A-PAGE). Four electrophoretic zones (I-IV) were identified and analyzed in the electrophoregrams (Fig 4).

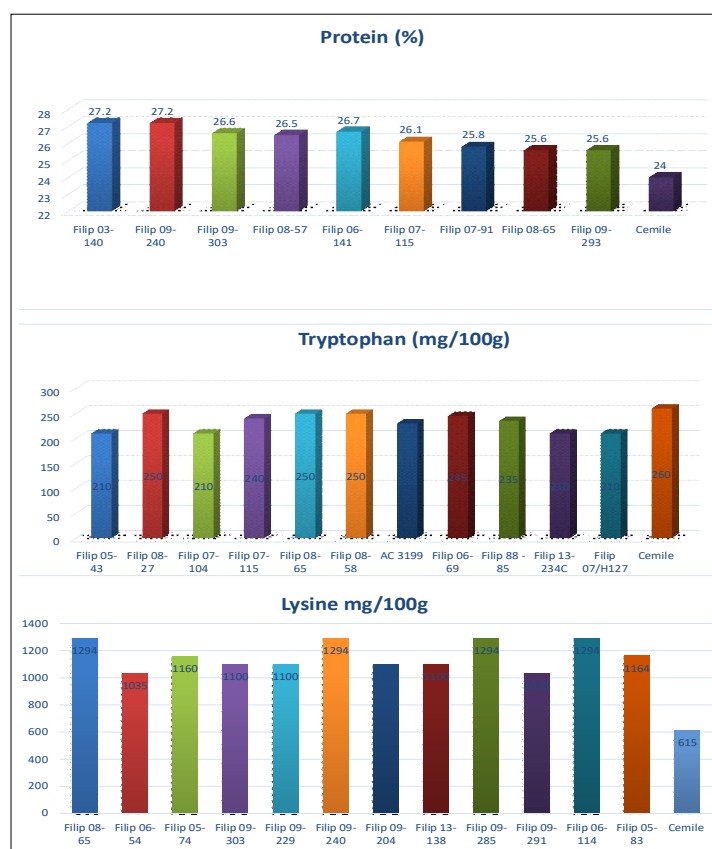


Fig 2: Biochemical composition of samples based on protein, tryptophan and lysine contents.

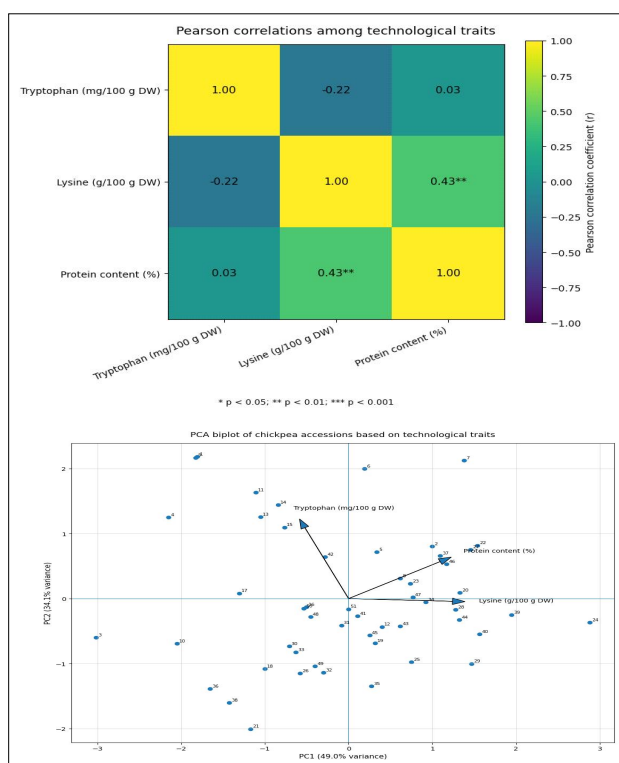


Fig 3: Biochemical trait relationships among chickpea accessions: Pearson correlation heatmap for protein, lysine and tryptophan contents and principal component analysis (PCA) biplot of the chickpea accessions based on the same biochemical traits.

Nei's diversity indices for electrophoretic storage-protein patterns were calculated for individual polymorphic zones of globulin storage protein electrophoregrams were

determined in the chickpea accessions ($H_{IV} = 0.464$, $H_{III} = 0.664$, $H_{II} = 0.752$ and $H_I = 0.562$). The highest storage-protein pattern diversity was observed in the β region ($H_{\beta} =$

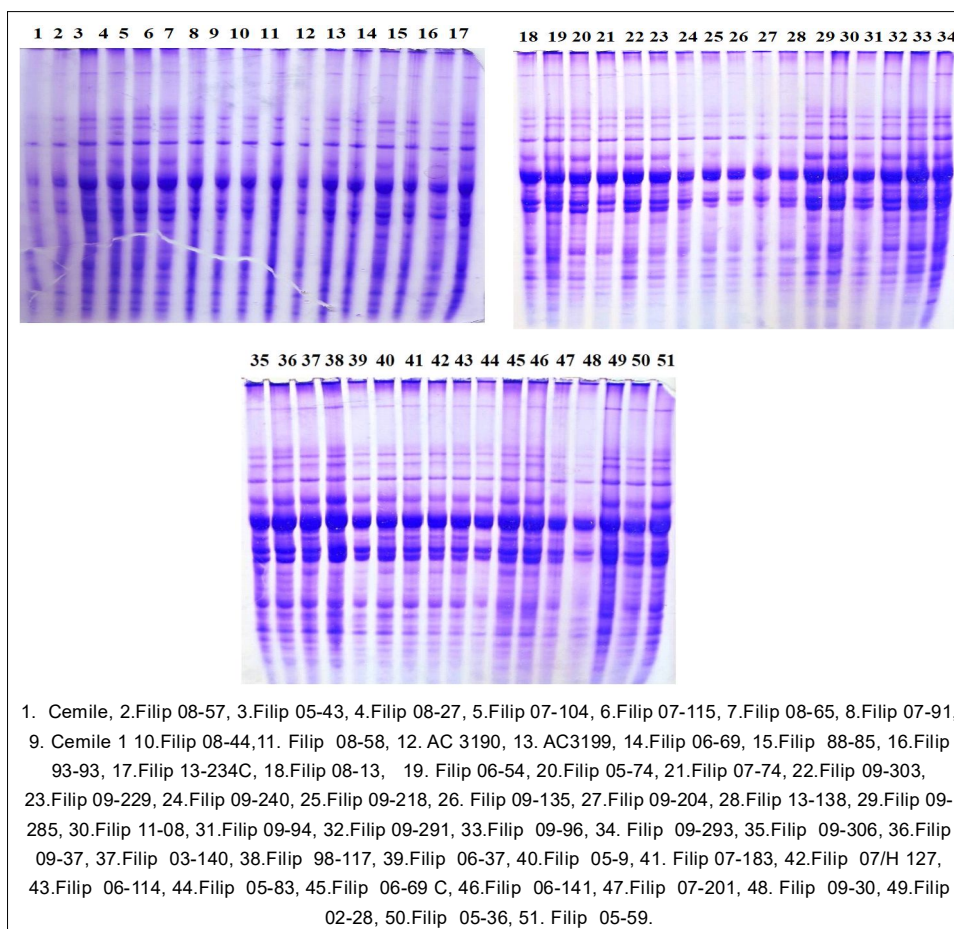


Fig 4: Electrophoretic analysis of globulin storage proteins in the seeds of chickpea (*Cicer arietinum* L.) accessions.

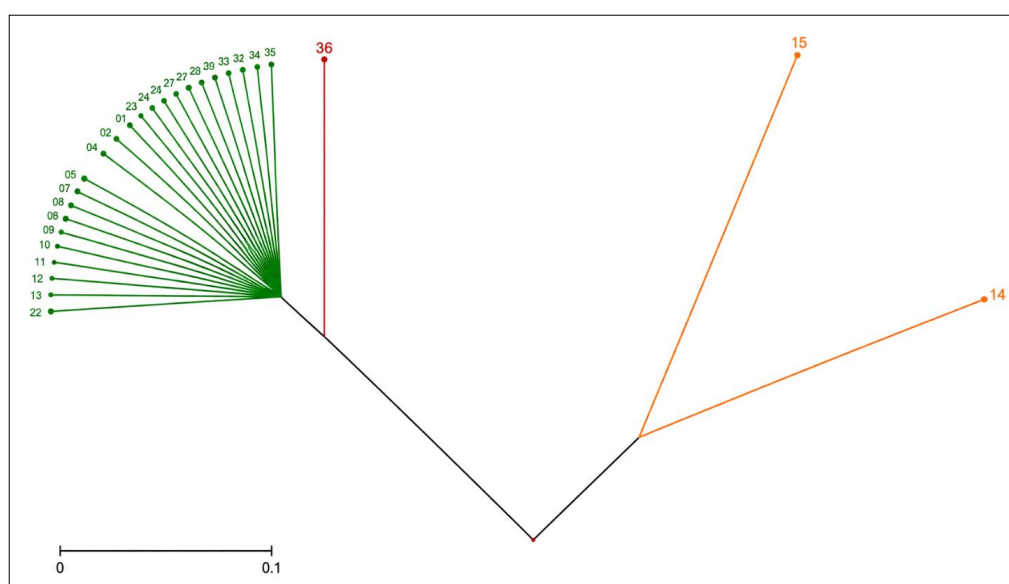


Fig 5: Results of the statistical analysis of electrophoregrams of chickpea accessions.

0.752). Compared with the other zones, the β zone exhibited greater variation in the number, intensity and migration positions of protein bands, indicating its important role in detecting genetic variation. In another study, genetic diversity was investigated in 76 chickpea accessions. Electrophoretic analysis showed that the γ and β regions had the greatest diagnostic value, enabling reliable differentiation among the accessions. These findings confirm that electrophoretic profiles of globulin proteins can serve as a reliable biochemical tool for assessing genetic diversity, identifying cultivars and selecting promising chickpea forms for future breeding programs (Salmanova, 2026).

Based on the dendrogram derived from globulin storage-protein electrophoretic profiles, the chickpea accessions were separated into four clusters (Fig 5). Accessions 14, 15 and 36 formed individual branches, indicating that their storage-protein profiles were distinct from those of the remaining accessions. Most accessions were grouped within one major cluster, reflecting comparatively similar electrophoretic banding patterns. These results represent biochemical differentiation based on storage-protein profiles and should not be interpreted as evidence of genome-wide genetic distinctness. Confirmation of genetic relationships requires DNA-based marker analysis, such as SSR or ISSR approaches (Bouri *et al.*, 2020; Janghel *et al.*, 2021).

CONCLUSION

Based on descriptive screening thresholds, nine accessions showed comparatively high protein contents ($\geq 25.6\%$ DW), 12 showed high lysine contents (≥ 1035 mg/100 g DW) and 11 exhibited high tryptophan contents (≥ 210 mg/100 g DW). Several accessions displayed favourable technological traits, although no multi-trait ranking was applied. Electrophoretic analysis of globulin storage proteins revealed considerable polymorphism among the 51 chickpea accessions, with the β region showing the highest storage-protein pattern diversity ($H\beta = 0.752$). These protein-profile differences represent biochemical variation and require validation through DNA-based molecular-marker analyses to determine their relationship with broader storage-protein pattern diversity.

The identified accessions with favourable nutritional characteristics and distinct storage-protein profiles may serve as preliminary candidates for chickpea breeding programmes aimed at improving seed quality. Further evaluation across environments and using molecular approaches is required to confirm their breeding potential.

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Disclaimers

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Conflict of interest

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